B-cell epitope prediction from antigen-only dynamics: project log

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Living record. Read top to bottom to get up to speed.

Running record of the project: predict B-cell epitope residues from *antigen-only* information, using alanine-scan $\Delta\Delta G_{\text{bind}}$ as residue-level binding-importance labels and asking whether antigen dynamics / solvation / energetics carry predictive signal *beyond static surface exposure*. Read in order; each section is one iteration — what we tried, what we found, what it changed. Not a formal manuscript. Every figure is regenerated by a named script (recorded in `figures/FIGURES.md`); every number traces to a file or command; unconfirmed values are flagged `[verify]`.

1 Motivation and the gap

We want to predict which residues of an antigen are B-cell epitope residues, from the antigen structure alone, in cases where the antibody is unknown. Doing this rigorously requires residue-level labels for "binding importance." We take these from **alanine scanning**: for residue i, $\Delta\Delta G^{(i)}_{\text{bind}} = \Delta G^{\text{mut},i}_{\text{bind}} - \Delta G^{\text{WT}}_{\text{bind}}$, with large positive values flagging residues whose wild-type identity stabilizes the interface. We treat this as a residue-level epitope score, $S_i^{\text{epitope}} \approx \Delta\Delta G^{(i)}_{\text{bind}}$. (Full derivation and the alchemical / StaB-ddG label tiers: `../../tcr/docs/b_cell_eptiope_prediction_may23_update_final.pdf`

The gap. Most B-cell epitope predictors label residues by *contact distance* to a bound antibody. That conflates "close" with "important": a residue can sit at the interface yet contribute little energetically, or be energetically critical without being the nearest neighbor. Alanine-scan $\Delta\Delta G$ is a cleaner, energetic definition of importance. Separately, since Westhof et al. [1] it has been known that antigenicity correlates with *segmental mobility* (flexibility / B-factor) — an antigen-intrinsic, antibody-agnostic property. The opportunity is to test, with modern MD, whether antigen-only *dynamical* and *solvation* features (RMSF, collective-mode participation, hydration thermodynamics) predict energetic binding importance — and whether they beat static exposure. This is where the author's methods background (FRESEAN collective-mode analysis; 3D-2PT-style hydration thermodynamics; non-equilibrium energy flow) is the differentiator (`../../_literature/my_manuscripts/`).

2 Design decisions to hold onto

Five issues decide whether the project produces a result or an artifact. Recorded here so they stay front-of-mind.

1. **The label is antibody-specific; the features are antibody-agnostic.** $\Delta\Delta G^{(i)}_{\text{bind}}$ is a residue's contribution to *one* paratope, but $\mathbf{x}_i$ comes from the apo antigen and cannot know

the antibody. So $f(\mathbf{x}_i) \to \Delta\Delta G^{(i)}$ can only recover the *antibody-marginal* (antigen-intrinsic) hotspot propensity. This is not a weakness to hide — it is exactly why dynamics/flexibility features should help (cf. Westhof). Where an antigen has several antibodies, average $\Delta\Delta G^{(i)}$ over them to *define* the intrinsic target.

2. **"Beyond SASA" is the primary hypothesis, not a footnote.** Surface exposure is the master confounder: exposed residues are simultaneously more likely to be hotspots, more flexible, and more hydrated. The publishable claim is not "flexibility predicts epitopes" (Westhof has that) but "dynamics/solvation features add signal *after* controlling for static exposure." Design: baseline = SASA + depth + residue type; then test the partial R^2 added by RMSF, mode participation, and 3D-2PT hydration.

3. **Effective $N \approx$ #antigens/folds, not #residues.** Residues within an antigen are correlated and the intrinsic signal is per-fold. Split **leave-one-complex-out** (ideally leave-one-fold-out); never random residue splits. At this scale the work is hypothesis-generating — powered for a strong dynamics-vs-exposure effect, not a deployable predictor. State this plainly.

4. **Features come from the apo antigen** — which is the point and sets a ceiling. Cryptic / induced-fit epitopes (conformational selection vs. induced fit) cannot be predicted in principle from apo features. Conformational-selection epitopes *should* show in apo flexibility/mode features; pure induced-fit ones will not. That contrast is itself a result.

5. **The MD protocol is not one stack.** A flexibility/RMSF run differs from a FRESEAN/3D-2PT run (the latter needs high-frequency velocity output, constraint removal, and a water model validated for dynamics). Likely two protocols per antigen. The house TIP3P/2 fs/LINCS default is scrutinized accordingly (see `../docs/METHODS.md`).

3 The label benchmark

Curated experimental alanine-scan $\Delta\Delta G$ for antigen-side residues lives in `benchmark/antigen_alanine_scanning` (6 sheets: README, Simple_Primary, Full_Schema [51 cols], hGH_Table2_4, Paper_Inventory, Counts). Audited 2026-06-18.

Coverage. 203 primary rows, 13 datasets, 7 antigens. Source: AB-Bind [2] (195 rows; `github.com/sarahsirin/` and the Jin 1994 hGH thesis (8 rows). Sign convention: $+\Delta\Delta G \Rightarrow$ weaker binding $\Rightarrow$ residue important (38 negative = binding-improving alanines; 17 zeros).

Integrity. The $\Delta\Delta G \leftrightarrow K_d$-ratio arithmetic is internally consistent for all 203 rows ($< 2\%$ vs $\exp(\Delta\Delta G/RT)$). No censoring in the primary set. The 4 non-alanine rows (hGH, EC50-derived) are correctly flagged `exclude_from_binding_label`. Provenance (DOI, source line, URL) is complete except the two homology-modeled BoNT/A2 sets and the Jin rows. `transcription_confidence` is uniformly `high` — treat as a soft flag, possibly a default fill. `[verify]` spot-check 1–2 known hotspots against the source papers.

Clean MD-correlation core. Dropping homology models (`HM_`) and EC50/excluded rows leaves **163 rows / 5 antigens on real PDBs** (VEGF, lysozyme, HPr, BoNT/A1, MT-SP1).

4 First system: hen egg-white lysozyme (HEL)

HEL is the cleanest entry point: small (129 residues), rigid (apo ≈ bound), very stable, textbook AMBER behavior, and — crucially — it carries **three** independent antibody alanine scans in the benchmark, on a consistent 1–129 numbering:

Antibody	PDB	Ag chain	# Ala	$\Delta\Delta G$ range (kcal/mol)
HyHEL-63	1DQJ	C	11	0.60–6.10
D1.3	1VFB	C	12	0.20–2.90
HyHEL-10	3HFM	Y	4	1.03–6.83

Why this is better than one antibody. HyHEL-10 and HyHEL-63 share an epitope and *agree* on hotspots (K96: 6.83 vs 6.10; Y20: 4.87 vs 3.30; K97: 6.18 vs 3.50; R21: 1.03 vs 1.30) — direct evidence for the antigen-intrinsic hotspot idea (Design note 1). D1.3 binds a *non-overlapping* epitope (residues 18–24 and 116–129; zero residue overlap with the HyHELs), giving a second epitope plus built-in negative controls. Because we extract features from the apo antigen, **one apo-HEL trajectory** serves all three antibodies and their mean.

Starting structure. RCSB **1AKI** (1.5 Å X-ray, chain A, apo). Verified: the 1AKI sequence matches the benchmark WT identity at all 23 labeled positions, so labels map 1:1 onto 1–129 numbering (K96/K97/Y20 hotspots confirmed). Apo crystal chosen over antigen-stripped-from-complex to match the prediction scenario; HEL's rigidity makes the choice low-risk.

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5 Simulated antigens

One section per antigen we simulate: what it is, the structural handle we use, and its alanine-scan labels. Structure-image placeholders render a box until the figure is added (see `figures/FIGURES.md`). $\Delta\Delta G_{\text{bind}}$ values are the change in binding free energy, K_d/K_i-derived; positive weakens binding.

5.1 Lysozyme (hen egg-white, HEL)

Hen egg-white lysozyme is a 129-residue glycoside hydrolase and the canonical model antigen of structural immunology: multiple high-resolution antibody complexes and alanine scans exist (HyHEL-10, HyHEL-63, D1.3). It is compact and conformationally rigid, cross-linked by four disulfides. We simulate the apo enzyme from `1AKI`. The labels here are the HyHEL-63 scan (`1DQJ`, Table 1); the v2 benchmark adds D1.3 (`1VFB`) and HyHEL-10 (`3HFM`) scans for the same antigen.

5.2 HPr

The histidine-containing phosphocarrier protein (HPr) is a small (∼9 kDa, 85-residue) cytoplasmic protein of the bacterial phosphoenolpyruvate:sugar phosphotransferase system, shuttling a phosphoryl group between enzyme I and the sugar-specific EIIA components. The *E. coli* protein is a

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Figure 1: Apo HEL (1AKI); label residues highlighted. *Placeholder.*

Table 1: **Lysozyme** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT \ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: apo 1AKI). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	HyHEL-63 (1DQJ)
Y20	+3.30
R21	+1.30
W62	+0.80
W63	+1.30
L75	+1.50
T89	+0.80
N93	+0.60
K96	+6.10
K97	+3.50
S100	+0.80
D101	+1.30

well-folded open-faced β-sandwich with *no cysteines*, making it an exceptionally clean dynamics test case. The Jel42 epitope clusters in the C-terminal half (Table 2). We simulate the apo protein from `2JEL` (chain P).

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Figure 2: Apo HPr (2JEL chain P). *Placeholder.*

5.3 VEGF-A

Vascular endothelial growth factor A is a secreted pro-angiogenic cytokine and a central drug target (bevacizumab, ranibizumab). Its receptor-binding domain is an antiparallel, disulfide-linked homodimer with a cystine-knot growth-factor fold; the two protomers are covalently joined, so the biological unit must be simulated as the dimer. The benchmark pairs two antibodies against the same receptor-binding face: fab-12 (`1BJ1`, the bevacizumab precursor) and the affinity-matured Y0317 (`1CZ8`, ranibizumab) — a useful same-antigen/different-affinity comparison (Table 3). We simulate the apo dimer from `1BJ1` (chains V, W).

Table 2: **HPr** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; competition, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 2JEL chain P). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	Jel42 (2JEL)
T62	+0.00
E68	+0.41
E70	+2.75
H76	-0.41
E83	+0.00
E85	+0.00

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Figure 3: Apo VEGF-A homodimer (1BJ1 chains V,W). *Placeholder.*

Table 3: **VEGF** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 1BJ1 chains V,W (dimer)). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	fab-12 (1BJ1)	Y0317 (1CZ8)
F17	+0.00	+0.00
Y21	+0.00	+0.00
Y45	+0.82	+1.94
K48	+0.41	+0.00
Q79	+0.00	+0.65
I80	+0.82	+0.95
M81	+3.69	+4.10
R82	+3.69	+0.82
I83	+3.69	+1.30
K84	+0.65	+1.36
H86	+0.00	+0.00
Q87	+0.00	+0.00
G88	+2.76	+2.67
Q89	+1.75	+1.06
H90	+0.00	+0.00
I91	+0.41	+1.06
G92	+3.69	+4.10
E93	+0.82	+1.15
M94	+1.42	+1.91

5.4 MT-SP1 (matriptase)

MT-SP1 / matriptase (ST14) is a type II transmembrane serine protease of the S1 family; its extracellular catalytic domain is over-expressed in epithelial cancers and is a validated oncology target. The benchmark uses two inhibitory Fabs, E2 (`3BN9`) and S4 (`3NPS`), scanning the protease domain in chymotrypsin numbering (Table 4). Simulation is staged but **pending**: the crystal structures lack two internal residues (149, 218) that must be loop-modelled before MD.

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Figure 4: MT-SP1 protease domain (3NPS chain A). *Placeholder.*

5.5 Bont/A1 (H_C receptor-binding domain)

Botulinum neurotoxin serotype A1 is among the most potent toxins known: a ∼150 kDa di-chain protein comprising a zinc-metalloprotease light chain and a heavy chain. Its C-terminal heavy-chain domain (H_C, here residues 872–1295) is the receptor-binding domain — engaging neuronal gangliosides and the SV2 protein — and the principal protective antigen targeted by neutralizing antibodies. We simulate the isolated H_C domain (`2NYY` chain A); the catalytic light-chain zinc lies outside this fragment. Two neutralizing mAbs, CR1 (`2NYY`) and AR2 (`2NZ9`), provide the labels (Table 5).

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Figure 5: Bont/A1 H_C domain (2NYY res 872–1295). *Placeholder.*

6 Methods — apo-HEL run #1

Frozen, signed-off stack in `../docs/METHODS.md` (2026-06-18). Purpose of run #1: *validate the prep→equil→production pipeline on Gemini and extract a first RMSF / flexibility feature* (the Westhof re-test). House default stack signed off as the starting point.

Stack. AMBER99SB-ILDN / TIP3P; cubic box, 1.2 nm clearance; neutralize + 0.15 M NaCl; leapfrog, 2 fs, LINCS on h-bonds; PME ($r_{\mathrm{coul}} = r_{\mathrm{vdw}} = 1.0$ nm, potential-shift, DispCorr); V-rescale at 300 K (Protein / Non-Protein); EM → 100 ps NVT → 100 ps NPT (both position-restrained) → **100 ns** production (single trajectory, for RMSF statistics). Random seeds (`gen_seed` = `ld_seed`

Table 4: **MT-SP1** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; enzymatic inhibition, K_i-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 3NPS chain A). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	E2 (3BN9)	S4 (3NPS)
Q23	–	+0.03
I26	–	+0.65
Q38	-0.42	–
I41	+0.00	–
I45	–	-0.34
D46	–	+0.34
D47	–	+1.08
R48	–	-1.07
F50	–	+0.22
R51	–	+0.14
Y52	–	+0.46
I60	+0.84	–
R82	–	-0.15
R87	-0.16	–
F89	–	+1.61
N90	–	+0.26
D91	–	+1.52
F92	–	+0.47
T93	–	+0.73
F94	+0.64	–
N95	+0.77	–
D96	+6.70	–
F97	+6.70	–
T98	+1.13	–
H138	–	+1.89
Q140	–	+0.30
Y141	–	+1.79
H143	+0.09	–
T144	–	+0.18
Q145	+0.13	–
Y146	+1.09	–
L147	–	+0.30
T150	+0.29	–
L153	+0.34	–
E163	–	+0.62
Q168	–	-0.06
E169	+0.37	+0.75
Q174	-0.03	–
Q175	+2.51	–
D214	–	+1.48
D217	+0.57	–
Q218	–	-0.04
R219	–	-0.08
K221	–	-0.10
R222	-0.09	–
K224	+0.79	–

Table 5: **Bont/A1** alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ (kcal/mol; KinExA, K_d-derived via $\Delta\Delta G_{\mathrm{bind}} = RT\ln(K^{\mathrm{mut}}/K^{\mathrm{wt}})$, $T = 310\,\mathrm{K}$; start structure: extract 2NYY Hc (res 872-1295)). Positive $\Delta\Delta G_{\mathrm{bind}}$ weakens binding.

Residue	CR1 (2NYY)	AR2 (2NZ9)
S902	-0.23	-0.12
K903	+0.29	+0.54
Q915	+0.52	+0.10
F917	+0.42	-0.05
N918	+0.89	+2.16
L919	+2.59	+2.28
E920	+2.84	+2.77
K923	-0.14	-0.30
F953	+4.06	+3.34
N954	+0.10	-0.15
S955	+0.00	-0.08
I956	-0.01	+0.07
K1056	-0.03	-0.01
D1058	+0.01	-0.03
R1061	+0.82	+0.29
D1062	+2.53	+2.34
T1063	+1.62	+2.37
H1064	+7.32	+7.42
R1294	+0.30	+0.39

$= -1$): bitwise reproducibility is not the target — the workflow reproduces from git-tracked inputs and saved derived outputs, and random seeds give genuinely independent replicas. Frozen per-run `.mdp` files: `md/1AKI/apo/configs/` (run dirs are organized `md/<PDB>/<state>/`).

Deliberate deviations from the house scripts. (i) **C-rescale** barostat for NPT equilibration (Parrinello-Rahman only in production), since PR can ring when started far from equilibrium; (ii) **cubic** box (the old house scripts disagreed cubic/dodecahedron) — chosen for interpretability and uniform across all systems, at a modest extra-solvent cost vs. dodecahedron.

HEL-specific checkpoints (not glossed over). pdb2gmx must form HEL's **4 disulfides** (6–127, 30–115, 64–80, 76–94) — verify the log reports 4 SS bonds. His15 protonation: record the assigned state. Crystal waters (78 HOH) removed; fresh TIP3P added. Read every grompp warning rather than letting `-maxwarn` hide it.

Caveats / flags. TIP3P over-mobilizes water and 2 fs+LINCS suppress H dynamics — fine for relative RMSF and pipeline validation, *not* for the dynamics-grade features. 100 ns is a single trajectory for statistics; replicas are deferred (random seeds make adding independent ones trivial).

Deferred: the distinctive run. FRESEAN mode participation and 3D-2PT hydration thermodynamics require a *separate* protocol — high-frequency velocity output, constraint removal / sub-fs dt, TIP4P/2005 — with settings pulled from the FRESEAN and energy-transfer manuscripts. This is the run that powers the mode-based and hydration feature classes.

7 Run log

Exact-command reproducibility ledger. Append a row per stage; pin the GROMACS version actually used on Gemini, seeds, and Slurm job id.

Date	Stage	Command	GROMACS ver.	Job / output
2026-06-18	structure	`curl .../1AKI.pdb`	—	`structures/1AKI.pdb`
pending	prep	`sbatch md/1AKI/apo/prep.sh`	2023.2-dev	out/npt.gro/cpt
pending	production	`sbatch md/1AKI/apo/md.sh`	2023.2-dev	out/md.xtc

Status (2026-06-18). Project scaffolded; benchmark audited; first system (HEL/1AKI) selected and label mapping verified 1:1; apo-HEL run #1 staged and signed off. Gemini environment confirmed: module `Gromacs` (capital G), GROMACS `2023.2-dev` (avx2_256), partition `gpu-a100` (4-day limit, A100 fastest); repo cloned to `/scratch/bneff/bcell_epitope`. Production walltime set to 48 h ceiling. **Next:** `git pull` on Gemini, then `sbatch prep.sh` → `sbatch md.sh`; record Slurm job ids and ns/day. Production length 100 ns (single trajectory). Open: replicas (deferred — random seeds make them trivial); scheduling the FRESEAN/3D-2PT run.

Run-script fixes (bringing up run #1 on Gemini). Three issues surfaced and were fixed while getting prep to run; recorded for reproducibility:

- **Module name.** Gemini's module is `Gromacs` (capital G); lowercase `gromacs` errors on a missing `shared` dependency. Scripts updated.
- **Config paths under Slurm.** Slurm copies the batch script to a private spool dir, so `$(dirname $0)/configs` resolved into the spool dir and `ions.mdp` was not found. Scripts now anchor paths to `$SLURM_SUBMIT_DIR`; **always submit from the repo root**.
- **GPU bonded offload on EM.** `-bonded gpu` requires a dynamical integrator; EM uses `steep`, so the EM `mdrun` now uses `-nb gpu` only. NVT/NPT/ production use `md` and keep `-nb/-pme/-bonded gpu` (audited).

Also removed the obsolete `ns_type` entry from all mdp files (ignored under the Verlet scheme). **Caveat:** Gemini scratch is not durable — move logs/outputs off scratch after each run (an earlier prep's logs were lost to a purge). Sanity check per run: `pdb2gmx.log` must report **4 disulfides** for HEL.

Expansion to multiple antigens (2026-06-21). Switched the box from dodecahedron to **cubic** (interpretability; uniform across systems) and re-ran HEL for consistency. Staged three further apo-antigen systems from the benchmark, hybrid-sourced (strip antigen chain(s) from the complex; numbering kept aligned to the labels). Provenance:

- **HPr** — `2JEL` chain P (res 1–85, no Cys); `md/2JEL/apo`.
- **VEGF** — `1BJ1` chains V+W (res 14–107, disulfide-linked homodimer; `pdb2gmx -merge all` for the inter-chain SS); `md/1BJ1/apo`.
- **Bont/A1 Hc** — `2NYY` chain A res 872–1295 (C-terminal receptor-binding domain only; catalytic-domain Zn/Ca excluded; res 872 is a domain cut → charged N-terminus); `md/2NYY/apo`.

All four (HEL+these) submitted overnight as prep→production chains (`sbatch --dependency=afterok`). **Deferred:** MT-SP1 (2 missing internal residues need loop modelling), Bont/A2 (homology models), hGH (provisional labels). Per-system disulfide expectations recorded in each `prep.sh`.

References

[1] E. Westhof, D. Altschuh, D. Moras, A. C. Bloomer, A. Mondragon, A. Klug, and M. H. V. Van Regenmortel. Correlation between segmental mobility and the location of antigenic determinants in proteins. *Nature*, 311:123–126, 1984.

[2] Sarah Sirin, James R. Apgar, Eric M. Bennett, and Amy E. Keating. Ab-bind: Antibody binding mutational database for computational affinity predictions. *Protein Science*, 25(2):393–409, 2016.